

ON THE COMPLEX VOLUME OF HYPERBOLIC KNOTS

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ABSTRACT

In this paper, we give a formula of the complex volumes of hyperbolic knot complements, which is related to the volume conjecture for hyperbolic knots.

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0. Introduction

Let M be the complement of a hyperbolic knot K in S^3 . In this paper, we call

$$\text{cv}(M) = -2\pi^2 \text{cs}(M) + \sqrt{-1} \text{vol}(M) \pmod{\pi^2}$$

the *complex volume* of M , where $\text{vol}(M)$ is the hyperbolic volume of M and $\text{cs}(M)$ is the Chern–Simons invariant of M defined in [3]. The purpose of this paper is to give an explicit formula of $\text{cv}(M)$ by applying the formula due to Neumann [6] and Zickert [10], which enables us to compute the complex volume of cusped hyperbolic 3-manifolds from their triangulations, to an ideal triangulation of M associated to a diagram D of K . Such a triangulation appears in the study of the *volume conjecture* [2, 4, 9]

$$\lim_{N \rightarrow \infty} \frac{2\pi}{N} \log |J_K(N; \exp 2\pi\sqrt{-1}/N)| = \text{vol}(M),$$

where $J_K(N; q)$ is the N -colored Jones polynomial of K , and our formula comes from an “optimistic” limit of $J_K(N; \exp 2\pi\sqrt{-1}/N)$ as expected in [5]. It should be noted that such a formula is already known for hyperbolic twist knots, see [1].

This paper is organized as follows. We associate D with an ideal triangulation of M in Sec. 1 and with a “potential” function, a sum of dilogarithm functions

$$\mathrm{Li}_2(x) = - \int_0^x \frac{\log(1-t)}{t} dt,$$

in Sec. 2 whose critical points correspond to the parabolic representations of $\pi_1(M)$ into $\mathrm{PSL}(2, \mathbb{C})$. Then, the complex volume of M is given as the critical value of this function corresponding to the complete hyperbolic structure of M , see Theorem 2.6. In Sec. 3, we briefly introduce Zickert’s formula in terms of our ideal triangulation of M . Then, we give a proof of Theorem 2.6 in Sec. 4, and give an example, the complex volume of the knot 8_{20} , in Sec. 5.

1. An Ideal Triangulation of M

In this section, we associate a diagram $D \subset S^2$ of K with an ideal triangulation $\mathcal{S}$ of M . Good references for such ideal triangulations and hyperbolicity equations of cusped hyperbolic 3-manifolds are [7, Sec. 2] and [8].

1.1. An ideal triangulation of $(S^2 \times \mathbb{R}) \setminus K$

In what follows, we identify S^3 with two poles, say P_+, P_- , removed with $S^2 \times \mathbb{R}$, and we consider D as a 4-valent graph in $\Sigma = S^2 \times \{0\}$. Let D^* denote the dual graph of D . Then, we may assume the following.

Assumption 1.1. There is no multiple edge in D^* .

Otherwise, we can reduce the number of crossings in D , say c . Let

$$\begin{aligned} \{X_n \mid 1 \leq n \leq c\}, \quad \{R_m \mid 0 \leq m \leq c+1\}, \\ \{P_l \mid 0 \leq l \leq 2c-1\}, \quad \{Q_g \mid 1 \leq g \leq 4c\}, \end{aligned}$$

denote the c crossings of D , the $c+2$ vertices of D^* , the $2c$ points in $D \cap D^*$ and the closures of the $4c$ connected components of $\Sigma \setminus (D \cup D^*)$, respectively. Furthermore,

$$\sigma(g) \in \{1, -1\}, \quad 1 \leq \nu(g) \leq c, \quad 0 \leq \mu(g) \leq c+1, \quad 0 \leq \alpha(g), \quad \beta(g) < 2c$$

are defined by Fig. 1, where $1 \leq g \leq 4c$.

We can suppose $K \subset S^2 \times [-1, 1]$, $K \cap \Sigma = D \cap D^*$ and

$$K \cap (S^2 \times \{\pm 1\}) = \{X_n \mid 1 \leq n \leq c\} \times \{\pm 1\},$$

respectively, where an arc component of $K_+ = K \cap (S^2 \times [0, 1])$ is called an *overpass* of K and an arc component of $K_- = K \cap (S^2 \times [-1, 0])$ is called an *underpass* of K . Obviously, $S^2 \times \mathbb{R}$ decomposes into $\{Q_g \times \mathbb{R} \mid 1 \leq g \leq 4c\}$, and each $Q_g \times \mathbb{R}$ looks like Fig. 2, where the bold arcs represent a part of an overpass, say K_{g+} , and a part

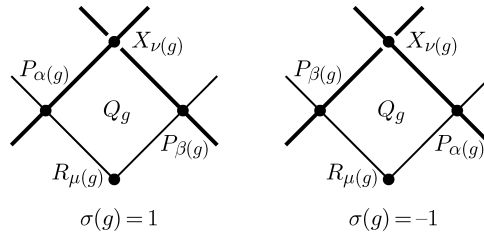


Fig. 1.

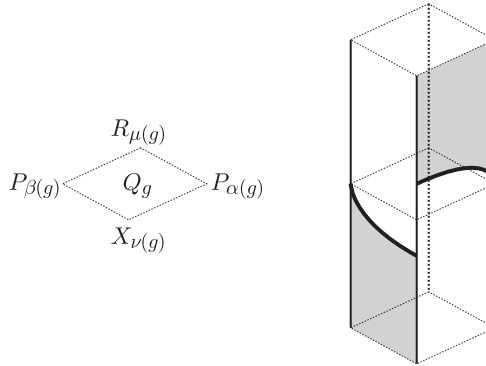


Fig. 2.

of an underpass, say K_{g-} , in $Q_g \times \mathbb{R}$. In what follows, for $1 \leq n \leq c$, $0 \leq m \leq c+1$ and $0 \leq l \leq 2c-1$, we put

$$A(X_n) = X_n \times (1, \infty), \quad B(X_n) = X_n \times (-\infty, -1), \quad C(X_n) = X_n \times (-1, 1),$$

$$A(P_l) = P_l \times (0, \infty), \quad B(P_l) = P_l \times (-\infty, 0), \quad C(R_m) = R_m \times \mathbb{R},$$

for simplicity. Let $A(K_{g+})$ be the face of $Q_g \times \mathbb{R}$ bounded by K_{g+} , $A(X_{\nu(g)})$, $A(P_{\alpha(g)})$, and $B(K_{g-})$ be the face of $Q_g \times \mathbb{R}$ bounded by K_{g-} , $B(X_{\nu(g)})$, $B(P_{\beta(g)})$, see Fig. 2.

Since each connected component of K_+ and K_- is contractible, we can collapse each connected component of $\bigcup_{g=1}^{4c} A(K_{g+})$ and $\bigcup_{g=1}^{4c} B(K_{g-})$ to a vertical edge. Then, as shown in Fig. 3, $Q_g \times \mathbb{R}$ becomes an ideal tetrahedron S_g in $(S^2 \times \mathbb{R}) \setminus K$ with four ordered ideal vertices and with six ideal edges

$$A(X_{\nu(g)}) = A(P_{\alpha(g)}), \quad A(P_{\beta(g)}), \quad B(X_{\nu(g)}) = B(P_{\beta(g)}), \quad B(P_{\alpha(g)}),$$

$$C(X_{\nu(g)}), \quad C(R_{\mu(g)}),$$

and we obtain an ideal triangulation $\mathcal{T}$ of $(S^2 \times \mathbb{R}) \setminus K$. In what follows, we denote the face of S_g opposite to the vertex k by $\partial_k S_g$, where $0 \leq k \leq 3$.

The set of edges from K to P_+ corresponds to the set of the connected components of K_+ , and the set of edges from K to P_- corresponds to the set of the

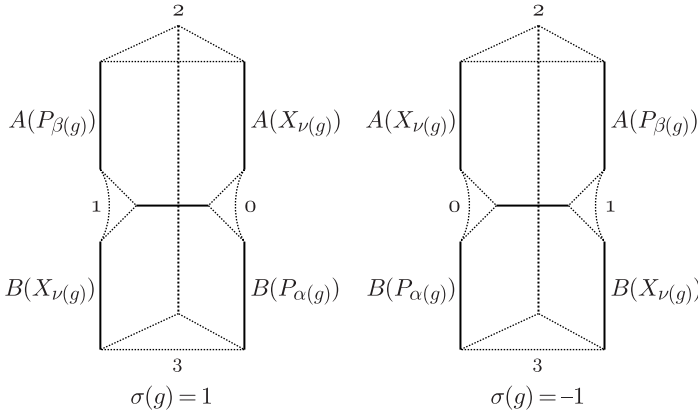


Fig. 3.

connected components of K_- . On the other hand, the set of edges from K to K corresponds to the set of crossings in D , and the set of edges from P_+ to P_- corresponds to the set of vertices of D^* .

1.2. An ideal triangulation of M

We choose a pair $\{I, J\}$ of an overpass I and an underpass J of K such that $I \cap J$ is one point, say P_0 . We can suppose that

$$I \cap \Sigma = \{P_0\} \cup \{P_l \mid 1 \leq l \leq a\}, \quad J \cap \Sigma = \{P_0\} \cup \{P_l \mid 2c - b \leq l < 2c\},$$

where $P_a \in \partial I$, $P_{2c-b} \in \partial J$, and that P_0 is on the edge of D^* incident to both R_0 and R_{c+1} . For simplicity, we put $\Lambda = \{l \in \mathbb{Z} \mid a < l < 2c - b\}$. Then, we may assume the following.

Assumption 1.2. The following subsets of $\{1, 2, \dots, 4c\}$ are mutually disjoint.

$$\begin{aligned} &\alpha^{-1}(\{1, \dots, a\}), \quad \beta^{-1}(a) \cup \alpha^{-1}(2c - b), \\ &\beta^{-1}(\{2c - b, \dots, 2c - 1\}), \quad \mu^{-1}(\{0, c + 1\}). \end{aligned}$$

Otherwise, we can reduce c . Furthermore, we may assume the following.

Assumption 1.3. $\beta^{-1}(a) \cap \alpha^{-1}(2c - b) = \emptyset$.

In fact, if there is no adjacent pair of an overpass and an underpass of K satisfying Assumption 1.3, K should be a link or the trefoil knot as shown in Fig. 4.

Then, we can suppose P_l is on the edge of D between X_l and X_{l+1} if $0 < l \leq a$, and on the edge between X_{l-c+1} and X_{l-c} if $2c - b \leq l < 2c$. Let

$$W = \bigcup_{\alpha(g)=0} (\partial_1 S_g) = \bigcup_{\beta(g)=0} (\partial_0 S_g)$$

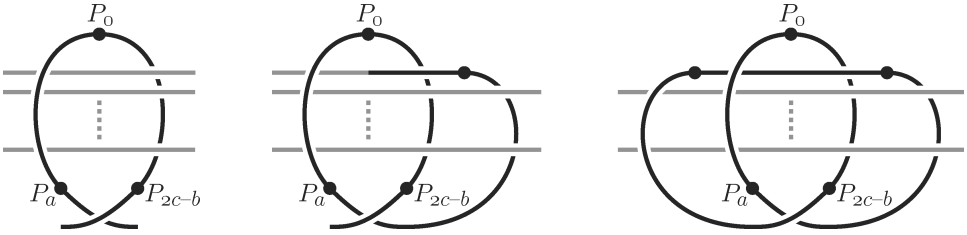


Fig. 4.

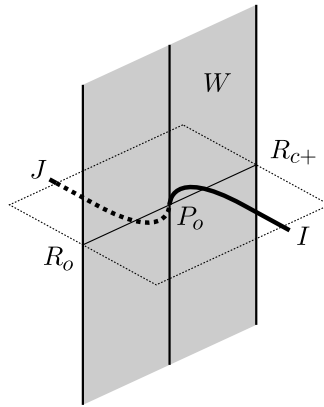


Fig. 5.

and $K' = K \cup W$, see Fig. 5. Then, M is homeomorphic to

$$M' = (S^2 \times \mathbb{R}) \setminus K',$$

and we can develop M' in the three-dimensional hyperbolic space $\mathbb{H}^3$.

Suppose $g \in \alpha^{-1}(\{1, \dots, a\}) \cup \beta^{-1}(a)$. Then,

$$A(P_0) = W \cap S_g = \begin{cases} A(X_{\nu(g)}) & \text{if } g \in \alpha^{-1}(\{1, \dots, a\}), \\ A(P_{\beta(g)}) & \text{if } g \in \beta^{-1}(a), \end{cases}$$

and we collapse $S_g \cap M'$ as shown in Fig. 6 because it will degenerate in $\mathbb{H}^3$ in this way. Note that

$$(C(X_{\nu(g)}), C(R_{\mu(g)})) = \begin{cases} (A(P_{\beta(g)}), B(P_{\alpha(g)})) & \text{if } g \in \alpha^{-1}(\{1, \dots, a\}), \\ (A(X_{\nu(g)}), B(X_{\nu(g)})) & \text{if } g \in \beta^{-1}(a). \end{cases}$$

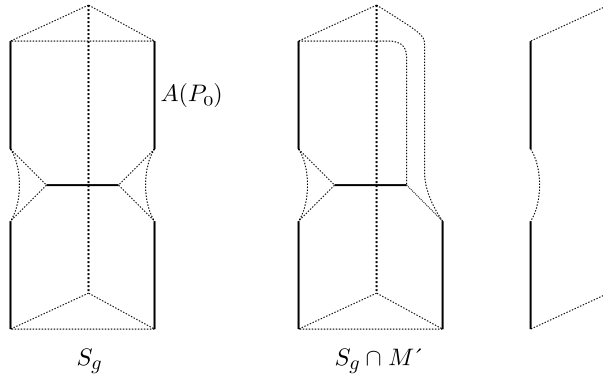


Fig. 6.

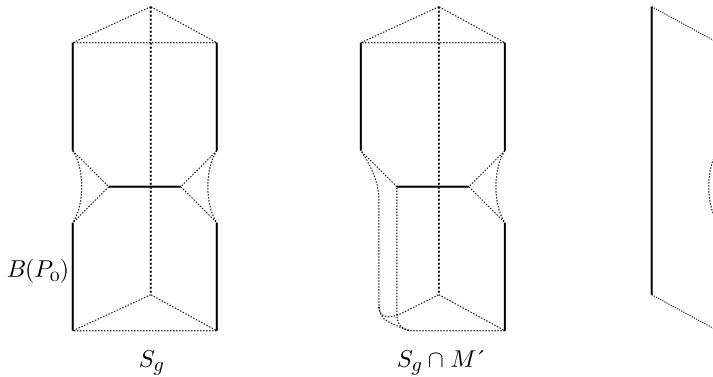


Fig. 7.

Suppose $g \in \alpha^{-1}(2c - b) \cup \beta^{-1}(\{2c - b, \dots, 2c - 1\})$. Then,

$$B(P_0) = W \cap S_g = \begin{cases} B(P_{\alpha(g)}) & \text{if } g \in \alpha^{-1}(2c - b), \\ B(X_{\nu(g)}) & \text{if } g \in \beta^{-1}(\{2c - b, \dots, 2c - 1\}), \end{cases}$$

and we collapse $S_g \cap M'$ as shown in Fig. 7. Note that

$$(C(X_{\nu(g)}), C(R_{\mu(g)})) = \begin{cases} (B(X_{\nu(g)}), A(X_{\nu(g)})) & \text{if } g \in \alpha^{-1}(2c - b), \\ (B(P_{\alpha(g)}), A(P_{\beta(g)})) & \text{if } g \in \beta^{-1}(\{2c - b, \dots, 2c - 1\}). \end{cases}$$

Suppose $g \in \mu^{-1}(\{0, c + 1\}) \setminus (\alpha^{-1}(0) \cup \beta^{-1}(0))$. Then, $W \cap S_g = C(R_{\mu(g)})$, and we collapse $S_g \cap M'$ as shown in Fig. 8. Note that

$$(A(X_{\nu(g)}), B(X_{\nu(g)})) = (B(P_{\alpha(g)}), A(P_{\beta(g)})).$$

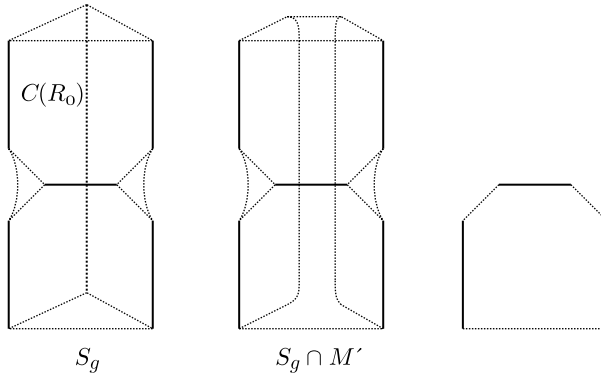


Fig. 8.

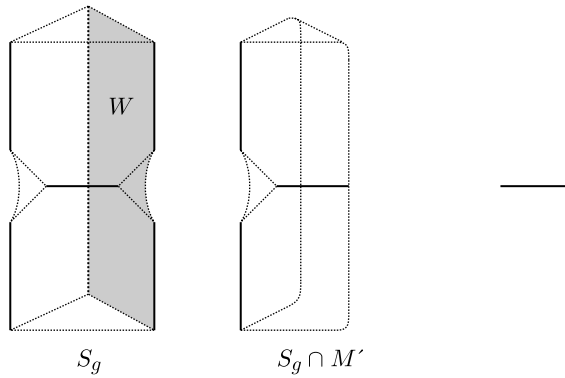


Fig. 9.

Suppose $g \in \alpha^{-1}(0) \cup \beta^{-1}(0)$. Then, $W \cap S_g$ is an ideal triangle in W , and we collapse $S_g \cap M'$ as shown in Fig. 9. Note that

$$C(X_{\nu(g)}) = \begin{cases} B(X_{\nu(g)}) = A(P_{\beta(g)}) & \text{if } g \in \alpha^{-1}(0), \\ A(X_{\nu(g)}) = B(P_{\alpha(g)}) & \text{if } g \in \beta^{-1}(0). \end{cases}$$

Let Γ denote $\{1, 2, \dots, 4c\}$ with

$$\alpha^{-1}(\{1, \dots, a\}), \quad \beta^{-1}(a) \cup \alpha^{-1}(2c-b), \quad \beta^{-1}(\{2c-b, \dots, 2c-1\}), \quad \mu^{-1}(\{0, c+1\})$$

removed. Then, S_g does not touch W if $g \in \Gamma$, and we stop collapsing. Now, we obtain an ideal triangulation $\mathcal{S}$ of M' , and so M .

1.3. A triangulation of a cusp cross section

In what follows, for a subset Y in S^3 , the interior of its regular neighborhood in S^3 is denoted by $N(Y)$. To write down the hyperbolicity equations for $\mathcal{S}$, we

will use the triangulation $\partial\mathcal{S}$ of $\partial N(K')$ induced by $\mathcal{S}$, and so we explain how to write down it. Let $\partial\mathcal{T}(P_+)$, $\partial\mathcal{T}(P_-)$ and $\partial\mathcal{T}(K)$ denote the triangulations of $\partial N(P_+)$, $\partial N(P_-)$ and $\partial N(K)$ induced by $\mathcal{T}$, respectively. If we denote the 2-simplex in them corresponding to the vertex k of S_g by $\Delta_k(S_g)$, we have

$$\Delta_0(S_g), \Delta_1(S_g) \in \partial\mathcal{T}(K), \quad \Delta_2(S_g) \in \partial\mathcal{T}(P_+), \quad \Delta_3(S_g) \in \partial\mathcal{T}(P_-).$$

Figure 10 describes how each $Q_g \times \mathbb{R}$ with $\nu(g) = n$ intersects $\partial N(P_+)$, $\partial N(P_-)$, $\partial N(K)$, and how each $\Delta_k(S_g)$, where $0 \leq k \leq 3$ and $\nu(g) = n$, looks like. This picture enables us to write down $\partial\mathcal{T}(P_+)$, $\partial\mathcal{T}(P_-)$, $\partial\mathcal{T}(K)$ piece by piece.

Let $O = \{P_+, P_-\} \cup C(R_0) \cup C(R_{c+1})$. First, the cell decomposition $\partial\mathcal{T}(O)$ of $\partial N(O)$ induced by $\mathcal{T}$ is obtained by gluing $\partial\mathcal{T}(P_+)$, $\partial\mathcal{T}(P_-)$ along $C(R_0) \cup C(R_{c+1})$ as shown in Fig. 11, where $\Delta_2(S_g), \Delta_3(S_g)$ are amalgamated to a quadrangle, say $\Delta_{23}(S_g)$, if $\mu(g) \in \{0, c+1\}$, see Fig. 8. Second, the cell decomposition $\partial\mathcal{T}$ of $S^3 \setminus N(K')$ induced by $\mathcal{T}$ is obtained by gluing $\partial\mathcal{T}(O)$, $\partial\mathcal{T}(K)$ along $W \cap (S^3 \setminus N(O \cup K))$, the gray annulus depicted in Fig. 11, which amalgamates

$$\begin{cases} \Delta_0(S_g), \Delta_{23}(S_g) & \text{if } g \in \alpha^{-1}(0), \\ \Delta_0(S_g), \Delta_2(S_g) & \text{if } g \in \alpha^{-1}(\{1, \dots, a\}), \\ \Delta_1(S_g), \Delta_2(S_g) & \text{if } g \in \beta^{-1}(a), \\ \Delta_0(S_g), \Delta_3(S_g) & \text{if } g \in \alpha^{-1}(2c-b), \\ \Delta_1(S_g), \Delta_3(S_g) & \text{if } g \in \beta^{-1}(\{2c-b, \dots, 2c-1\}), \\ \Delta_1(S_g), \Delta_{23}(S_g) & \text{if } g \in \beta^{-1}(0), \end{cases}$$

as shown in Figs. 6, 7 and 9. Finally, by collapsing a number of quadrangles (see Figs. 6–8) and eight special triangles (see Fig. 9) in $\partial\mathcal{T}$, we obtain $\partial\mathcal{S}$.

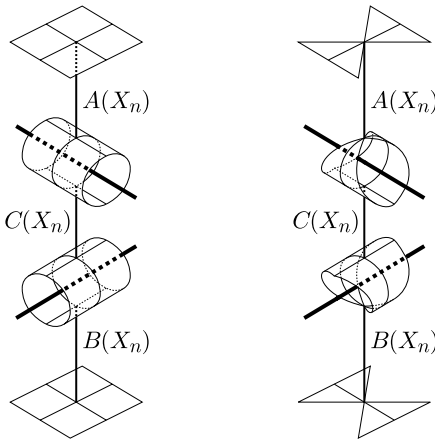


Fig. 10.

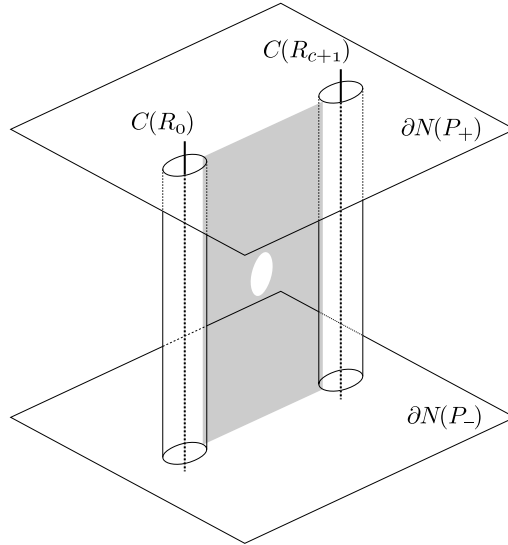


Fig. 11.

1.4. The edges in $\mathcal{S}$

Let G denote D with the edges intersecting I or J removed and $F_0, F_1, \dots, F_{c-a-b}$ the connected components of Σ with G removed, where we suppose $P_0 \in F_0, P_a \in F_p, P_{2c-b} \in F_q$. First of all, we may assume the following.

Assumption 1.4. $\#(I \cap F_f), \#(J \cap F_f) \leq 1$ and F_f is simply connected if $f \neq 0$.

Otherwise, we can reduce c or K is not prime. Then, for $f \neq 0$, the edges in

$$\{C(R_m) \mid R_m \subset F_f\} \cup \{B(P_l) \mid P_l \in I \cap F_f\} \cup \{A(P_l) \mid P_l \in J \cap F_f\}$$

become an edge, say $C(F_f)$, of $\mathcal{S}$ by the observations in Sec. 1.2. For simplicity, we put

$$\mathcal{C} = \{C(F_f) \mid 1 \leq f \leq c - a - b\} \cup \{C(X_n) \mid a + 1 \leq n \leq c - b\}.$$

Let $\{I_1, I_2, \dots, I_r\}$, and $\{J_1, J_2, \dots, J_s\}$ be the sets of connected components of

$$K_+ \cup \left(\bigcup_{n=1}^a \bigcup_{\nu(g)=n} K_{g-} \right) \quad \text{and} \quad K_- \cup \left(\bigcup_{n=c-b+1}^c \bigcup_{\nu(g)=n} K_{g+} \right),$$

respectively. Then, by the observations in Sec. 1.2, the edges in

$$\{A(P_l) \mid P_l \in I_i\} \cup \{A(X_n) \mid X_n \times \{1\} \in I_i\} \cup \{C(X_n) \mid X_n \times \{-1\} \in I_i\}$$

become an edge, say $A(I_i)$, of $\mathcal{S}$ for $1 \leq i \leq r$, and the edges in

$$\{B(P_l) \mid P_l \in J_j\} \cup \{B(X_n) \mid X_n \times \{-1\} \in J_j\} \cup \{C(X_n) \mid X_n \times \{1\} \in J_j\}$$

become an edge, say $B(J_j)$, of $\mathcal{S}$ for $1 \leq j \leq s$. For simplicity, we put

$$\mathcal{A} = \{A(I_i) \mid 1 \leq i \leq r\}, \quad \mathcal{B} = \{B(J_j) \mid 1 \leq j \leq s\}.$$

Then, the set of edges of $\mathcal{S}$ is given by $\mathcal{A} \cup \mathcal{B} \cup \mathcal{C}$, where

$$\begin{aligned} \mathcal{A} \cap \mathcal{B} &= \{A(I_i) = B(J_j) \mid I_i \cap J_j \cap \partial F_0 \neq \emptyset\}, \\ \mathcal{A} \cap \mathcal{C} &= \{A(X_{a+1}) = C(X_{a+1}), A(X_{c-b}) = C(F_q)\}, \\ \mathcal{B} \cap \mathcal{C} &= \{B(X_{c-b}) = C(X_{c-b}), B(X_{a+1}) = C(F_p)\}. \end{aligned}$$

2. Hyperbolicity Equations for M

As is well known, the hyperbolic structure of each S_g in $\mathcal{S}$ is determined by a complex number, called *modulus*, which is associated to a pair of opposite edges. In what follows, for $g \in \Gamma$, we associate z_g , $(1 - z_g^{-\sigma(g)})^{\sigma(g)}$, $(1 - z_g^{\sigma(g)})^{-\sigma(g)} \in \mathbb{C}$ to

$$\{C(X_{\nu(g)}), C(R_{\mu(g)})\}, \quad \{A(X_{\nu(g)}), B(X_{\nu(g)})\}, \quad \{A(P_{\beta(g)}), B(P_{\alpha(g)})\},$$

respectively. If such structures are extended to a hyperbolic structure of M , the moduli z_g s must satisfy the *edge relations*, that is, the product of the moduli around each edge of $\mathcal{S}$ should be 1. Furthermore, if the hyperbolic structure of M is complete, z_g s must satisfy the *cuspid condition*, that is, the product of the moduli along a meridian of K should be 1. The purpose of this section is to write down such equations explicitly.

2.1. Relations around the edges in $\mathcal{C}$

Define $\phi : \{1, 2, \dots, c\} \rightarrow \{1, 2, \dots, c - a - b\}$ by $R_m \subset F_{\phi(m)}$. Then, if $f \notin \{p, q\}$, the edge relation around $C(F_f)$ is given by

$$\prod_{\phi(\mu(g))=f} z_g = 1. \quad (2.1)$$

On the other hand, if $a + 1 < n < c - b$, the edge relation around $C(X_n)$ is given by

$$\prod_{\nu(g)=n} z_g = 1. \quad (2.2)$$

In what follows, we put

$$T_0 = \bigcup_{g \in \Gamma} (\Delta_0(S_g) \cup \Delta_1(S_g)), \quad T_+ = \bigcup_{g \in \Gamma} \Delta_2(S_g), \quad T_- = \bigcup_{g \in \Gamma} \Delta_3(S_g)$$

and denote the meridians $T_0 \cap T_{\pm}$ by $\Omega_{\pm}$ respectively.

Lemma 2.1. *The cuspid conditions along the annuli $N(\Omega_{\pm}) \cap T_{\pm}$ are given by*

$$\prod_{\phi(\mu(g))=p} z_g = \prod_{\nu(g)=a+1} z_g, \quad \prod_{\phi(\mu(g))=q} z_g = \prod_{\nu(g)=c-b} z_g, \quad (2.3a)$$

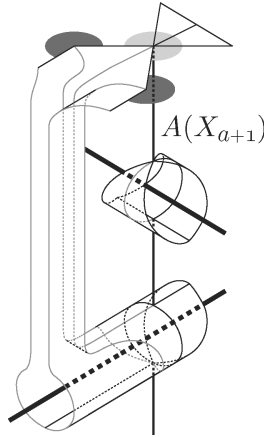


Fig. 12.

respectively, if $p \neq q$, modulo the edge relations around $C(X_{a+1})$ and $C(X_{c-b})$, and

$$\prod_{\phi(\mu(g))=p} z_g = \prod_{\nu(g) \in \{a+1, c-b\}} z_g \quad (2.3b)$$

if $p = q$, modulo the edge relations around $C(X_{a+1})$ and $C(X_{c-b})$ together with the cusp conditions along the annuli $N(\Omega_{\pm}) \cap T_0$.

Proof. If $p \neq q$, $\partial\mathcal{T}$ around $A(X_{a+1}) = C(X_{a+1})$ looks like Fig. 12, where the gray edges will be contracted in $\partial\mathcal{S}$, then the edge relation around $A(X_{a+1})$ equates the product of the moduli in the light-gray sector to $\prod_{\nu(g)=a+1} z_g^{-1}$, and the product of the moduli in the dark-gray sectors is equal to $\prod_{\phi(\mu(g))=p} z_g$. Therefore, the product π_+ of the moduli along the annulus $N(\Omega_+) \cap T_+$ is given by

$$\pi_+ = \prod_{\phi(\mu(g))=p} z_g \prod_{\nu(g)=a+1} z_g^{-1}.$$

We can compute the product π_- of the moduli along the annulus $N(\Omega_-) \cap T_-$ similarly.

If $p = q$, $\partial\mathcal{T}$ around $A(X_{a+1}) = C(X_{a+1})$ and $A(X_{c-b}) = C(F_p)$ looks like Fig. 13, then the cusp condition along the annulus $N(\Omega_-) \cap T_0$ equates the product of the moduli in the light-gray sector to $\prod_{\nu(g)=c-b} z_g^{-1}$, and the product of the moduli in the dark-gray sectors is equal to $\prod_{\phi(\mu(g))=p} z_g \prod_{\nu(g)=a+1} z_g^{-1}$ as above. Therefore,

$$\pi_+ = \prod_{\phi(\mu(g))=p} z_g \prod_{\nu(g) \in \{a+1, c-b\}} z_g^{-1}.$$

We can compute π_- similarly. □

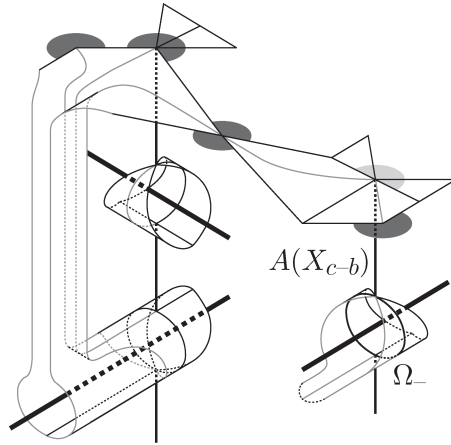


Fig. 13.

2.2. Relations around the edges in $\mathcal{A} \cup \mathcal{B}$

Let

$$E_1, \dots, E_d, E_{d+1}, \dots, E_{2(c-a-b)-1}$$

denote the connected components of G with $\{X_n \mid a < n \leq c - b\}$ removed, where we suppose $E_e \cap \partial F_0 = \emptyset$ if and only if $1 \leq e \leq d$. Then, we define

$$\epsilon : \Lambda \rightarrow \{1, \dots, d, d+1, \dots, 2(c-a-b)-1\}$$

by $P_l \in E_{\epsilon(l)}$ and, for $1 \leq e \leq d$, we put

$$\pi(e) = \prod_{\epsilon(\alpha(g))=e} (1 - z_g^{\sigma(g)})^{\sigma(g)} \prod_{\epsilon(\beta(g))=e} (1 - z_g^{\sigma(g)})^{-\sigma(g)}.$$

Then, the hyperbolicity equations for M other than (2.1)–(2.3a,b) can be described in terms of $\pi(e)$ s as follows. Suppose E_e with $1 \leq e \leq d$ is surrounded by Q_g, Q_h, Q_i, Q_j with $g, h, i, j \in \Gamma \cup \nu^{-1}(\{a+1, c-b\})$ as shown in Fig. 14.

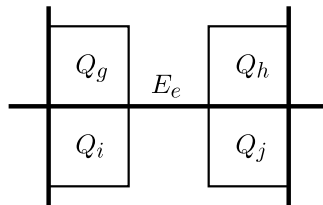


Fig. 14.

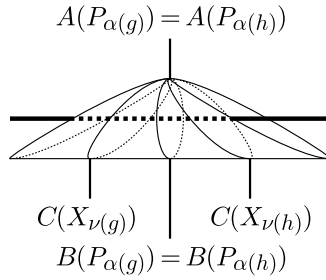


Fig. 15.

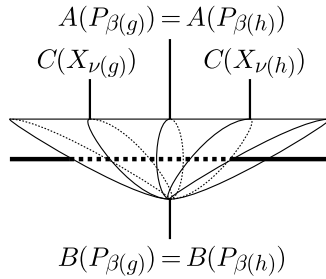


Fig. 16.

If $\alpha(g) = \alpha(i)$ and $\alpha(h) = \alpha(j)$, then T_0 around

$$B(P_{\alpha(g)}) = B(P_{\alpha(h)}) = B(P_{\alpha(i)}) = B(P_{\alpha(j)})$$

looks like Fig. 15, and the edge relation around it is given by $\pi(e) = 1$. Similarly, if $\beta(g) = \beta(i)$ and $\beta(h) = \beta(j)$, then T_0 around

$$A(P_{\beta(g)}) = A(P_{\beta(h)}) = A(P_{\beta(i)}) = A(P_{\beta(j)})$$

looks like Fig. 16, and the edge relation around it is given by $\pi(e) = 1$.

On the other hand, if $\alpha(g) = \alpha(i)$ and $\beta(h) = \beta(j)$, then T_0 around $C(X_{\nu(g)}), C(X_{\nu(h)})$ looks like Fig. 17, and the cusp condition along the annulus

$$\Delta_0(S_g) \cup \Delta_1(S_h) \cup \Delta_0(S_i) \cup \Delta_1(S_j)$$

is given by $\pi(e) = 1$. In what follows, we suppose

$$\dim(I_i \cap K_+) = \begin{cases} 1 & \text{if } 1 \leq i \leq t, \\ 0 & \text{if } t < i \leq r, \end{cases} \quad \dim(J_j \cap K_-) = \begin{cases} 1 & \text{if } 1 \leq j \leq t, \\ 0 & \text{if } t < j \leq s. \end{cases}$$

Then, we may assume the following.

Assumption 2.2. For $1 \leq i, j \leq t$, $\#(I_i \cap \partial F_0), \#(J_j \cap \partial F_0) \leq 1$.

Otherwise, we can reduce c .

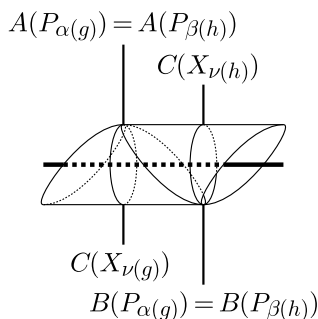


Fig. 17.

Proposition 2.3. *The hyperbolicity equations for M associated to $\mathcal{S}$ are given by*

$$\pi(1) = 1, \quad \pi(2) = 1, \dots, \pi(d) = 1 \quad (2.4)$$

together with (2.1)–(2.3).

Proof. Let $\pi(I_i)$ and $\pi(J_j)$ be the products of the moduli around $A(I_i)$ and $B(J_j)$, respectively. It suffices to show $\pi(I_i), \pi(J_j) = 1$ by using (2.1)–(2.4). We use the conventions $\pi(\epsilon(a)) = \pi_+$ and $\pi(\epsilon(2c - b)) = \pi_-$. If $t < i \leq r$, then $\pi(I_i) = 1$ is already contained in (2.4), and we suppose $1 \leq i \leq t$. If $\partial I_i \cap \partial F_0 = \emptyset$, then

$$\pi(I_i) = \pi(\epsilon(l))\pi(\epsilon(k))$$

modulo (2.2), (2.4), where we put $\partial I_i = \{P_l, P_k\}$, and the right-hand side is 1 by (2.3), (2.4). If $\partial I_i \cap \partial F_0 \neq \emptyset$, there uniquely exists $1 \leq j \leq t$ such that $\partial I_i \cap \partial J_j \cap \partial F_0 \neq \emptyset$, and

$$\pi(I_i) = \pi(\epsilon(l))/\pi(\epsilon(k)) = \pi(J_j)$$

modulo (2.2), (2.4), where we put $\partial(I_i \cup J_j) = \{P_l, P_k\}$, and this is equal to 1 by (2.3), (2.4). Similarly, we can prove $\pi(J_j) = 1$ for $1 \leq j \leq s$. $\square$

Remark 2.4. A solution to (2.1)–(2.4) is called *essential* if $z_g \notin \{0, 1, \infty\}$ for $g \in \Gamma$. If there exists such a solution, we can confirm the tetrahedra in $\mathcal{S}$ are nondegenerate, and the solutions correspond to the parabolic representations of $\pi_1(M)$ into $\mathrm{PSL}(2, \mathbb{C})$.

2.3. Potential function

There is a one-to-one correspondence between the set of solutions to (2.1)–(2.3) and the set of maps $z : \{1, 2, \dots, d\} \rightarrow \mathbb{C}$. In fact,

$$z_g^{\sigma(g)} = z(\epsilon(\beta(g)))/z(\epsilon(\alpha(g))),$$

where we put $z(e) = 1$ if $e > d$, gives such a correspondence. If we consider

$$V(z; D) = \sum_{g \in \Gamma} \sigma(g) \text{Li}_2(z_g^{\sigma(g)}),$$

as a function of $z(1), z(2), \dots, z(d)$, a simple calculation shows

$$\pi(e) = \exp \left\{ z(e) \frac{\partial V(z; D)}{\partial z(e)} \right\}$$

and so Proposition 2.3 can be stated as follows.

Theorem 2.5. *The hyperbolicity equations for M associated to $\mathcal{S}$ are given by*

$$z(e) \frac{\partial V(z; D)}{\partial z(e)} \in 2\pi\sqrt{-1}\mathbb{Z},$$

where D is a diagram of K satisfying Assumptions 1.1–1.4 and 2.2.

The following is the main result of this paper.

Theorem 2.6. *Suppose that the equation in Theorem 2.5 has an essential solution and that $\zeta : \{1, 2, \dots, d\} \rightarrow \mathbb{C}$ is the solution corresponding to the complete hyperbolic structure of M . Then, $\text{cv}(M)$ is equal to*

$$\hat{V}(\zeta; D) = V(\zeta; D) - \sum_{e=1}^d \log \zeta(e) \left[z(e) \frac{\partial V(z; D)}{\partial z(e)} \right]_{z=\zeta}.$$

3. Zickert's Formula

To apply Zickert's formula to $\mathcal{S}$, the orientations of the edges of each S_g induced by the vertex ordering should be consistent globally. However, the orientations of $A(X_{c-b})$ and $B(X_{c-b})$ do not agree with those of $C(F_q)$ and $C(X_{c-b})$ respectively, and so we put two ideal tetrahedra S_{4c+1} and S_{4c+2} between $C(X_{c-b})$ and $B(X_{c-b})$ and two ideal tetrahedra S_{4c+3} and S_{4c+4} between $A(X_{c-b})$ and $C(F_q)$ as shown in Fig. 18. The moduli z_{4c+1}, z_{4c+2} of S_{4c+1}, S_{4c+2} at the edge U_+ and the moduli z_{4c+3}, z_{4c+4} of S_{4c+3}, S_{4c+4} at the edge U_- are uniquely determined, and their vertex orderings are given by the edge orientations in Fig. 18. Furthermore, we put $\sigma(4c+1) = \sigma(4c+2) = -\sigma(4c+3) = -\sigma(4c+4) = 1$.

A Euclidean structure of $T = T_0 \cup T_+ \cup T_-$ associates an oriented edge y of $\partial\mathcal{S}$ with a vector in $\mathbb{E}^2$ represented by $\psi(y) \in \mathbb{C}$. Let x be an oriented edge in $\mathcal{S}$. Let y_+ and y_- be the oriented edges of $\partial\mathcal{S}$ depicted in Fig. 19, where Z is an ideal triangle in $\mathcal{S}$ containing x , and define $\xi(x)$ as a complex number satisfying $\xi(x)^{-2} = \psi(y_+)\psi(y_-)$. Then, $\xi(x)$ does not depend on the choice of Z , and we may assume that $\xi(x)$ does not depend on the orientation of x . Let x_{ij} denote the edge between the ideal vertices i and j of S_g and

$$u(S_g) = \log \xi(x_{03}) - \log \xi(x_{13}) + \log \xi(x_{12}) - \log \xi(x_{02}),$$

$$v(S_g) = \log \xi(x_{02}) - \log \xi(x_{23}) + \log \xi(x_{13}) - \log \xi(x_{01}).$$

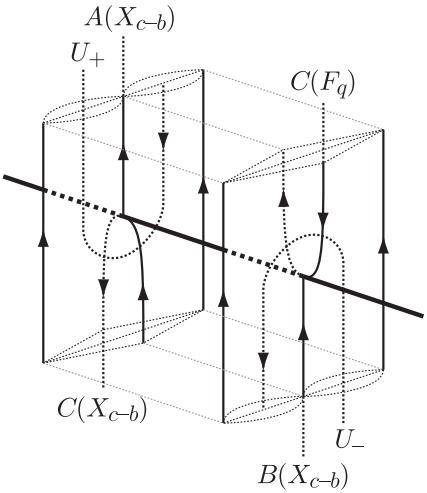


Fig. 18.

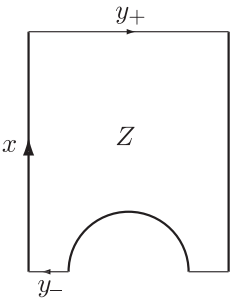


Fig. 19.

By [10, Lemma 3.15], there exist $\kappa_g, \lambda_g \in \mathbb{Z}$ such that

$$u(S_g) = \log z_g^{\sigma(g)} + \kappa_g \pi \sqrt{-1}, \quad v(S_g) = -\log(1 - z_g^{\sigma(g)}) + \lambda_g \pi \sqrt{-1}.$$

Now, by Zickert's formula [10], we have

$$\text{cv}(M) = \sum_{g \in \Gamma} \sigma(g) L(g) + \sum_{g=4c+1}^{4c+4} \sigma(g) L(g), \tag{3.1}$$

where $L(g)$ is defined by

$$\begin{aligned} L(g) &= \text{Li}_2(z_g^{\sigma(g)}) + \frac{1}{2} \log z_g^{\sigma(g)} \log(1 - z_g^{\sigma(g)}) - \frac{\pi^2}{6} \\ &\quad + \frac{\pi \sqrt{-1}}{2} \{ \lambda_g \log z_g^{\sigma(g)} + \kappa_g \log(1 - z_g^{\sigma(g)}) \}. \end{aligned}$$

Lemma 3.1. For any $l \in \Lambda$, $\xi(A(P_l))/\xi(B(P_l)) = \zeta(\epsilon(l))$.

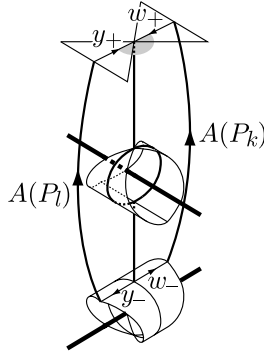


Fig. 20.

Proof. Suppose $a + 1 < n \leq c - b$ and $\beta(\nu^{-1}(n)) = \{l, k\}$. Let y_+, y_-, w_+, w_- denote the oriented edges of ∂S depicted in Fig. 20. Then, by definition,

$$\xi(A(P_l))^{-2} = \psi(y_+)\psi(y_-), \quad \xi(A(P_k))^{-2} = \psi(w_+)\psi(w_-).$$

On the other hand, as in the proof of Lemma 2.1, the cusp condition along the bold circle equates $\zeta(\epsilon(l))/\zeta(\epsilon(k))$ to the product of the moduli in the gray sector, and

$$\psi(w_+)/\psi(y_+) = \zeta(\epsilon(l))/\zeta(\epsilon(k)) = \psi(w_-)/\psi(y_-).$$

Therefore, we have $\xi(A(P_l))/\xi(A(P_k)) = \zeta(\epsilon(l))/\zeta(\epsilon(k))$, that is,

$$\frac{\xi(A(P_l))/\xi(B(P_l))}{\xi(A(P_k))/\xi(B(P_k))} = \zeta(\epsilon(l))/\zeta(\epsilon(k)).$$

Similarly, if we suppose $a + 1 \leq n < c - b$ and $\alpha(\nu^{-1}(n)) = \{l, k\}$, we can prove

$$\frac{\xi(A(P_l))/\xi(B(P_l))}{\xi(A(P_k))/\xi(B(P_k))} = \zeta(\epsilon(l))/\zeta(\epsilon(k)).$$

This implies Lemma 3.1 because $\xi(A(P_l))/\xi(B(P_l)) = 1$ if $P_l \in \partial F_0$. $\square$

Corollary 3.2. Suppose $g \in \Gamma$ or $4c + 1 \leq g \leq 4c + 4$. Then, κ_g is even.

Proof. Suppose $g \in \Gamma$. Then, by definition,

$$u(S_g) = \log \xi(B(P_{\alpha(g)})) - \log \xi(B(X_{\nu(g)})) + \log \xi(A(P_{\beta(g)})) - \log \xi(A(X_{\nu(g)})).$$

Since $A(X_{\nu(g)}) = A(P_{\alpha(g)})$ and $B(X_{\nu(g)}) = B(P_{\beta(g)})$, the right-hand side is equal to

$$-\log \zeta(\epsilon(\alpha(g))) + \log \zeta(\epsilon(\beta(g))) \equiv \log z_g^{\sigma(g)} \pmod{2\pi\sqrt{-1}\mathbb{Z}}$$

by Lemma 3.1. Suppose $4c + 1 \leq g \leq 4c + 4$ and $\beta(\nu^{-1}(c - b)) = \{l, k\}$, where

$$\zeta(\epsilon(l))/\zeta(\epsilon(k)) = \prod_{\nu(g)=c-b} z_g.$$

Note that the right-hand side is equal to z_{4c+1}^{-1} by the edge relation around $C(X_{c-b})$ and to z_{4c+4}^{-1} by (2.3a,b) and the edge relation around $C(F_g)$. Note also that $z_{4c+2} =$

z_{4c+1}^{-1} by the edge relation around U_+ and that $z_{4c+3} = z_{4c+4}^{-1}$ by the edge relation around U_- . Then, by definition and by the proof of Lemma 3.1, $(-1)^g u(S_g)$ is equal to

$$\log \xi(A(P_l)) - \log \xi(A(P_k)) \equiv \log \zeta(\epsilon(l)) / \zeta(\epsilon(k)) = (-1)^g \log z_g^{\sigma(g)} \pmod{2\pi\sqrt{-1}\mathbb{Z}}.$$

This completes the proof. $\square$

By Corollary 3.2, $L(g)$ is equal to

$$\begin{aligned} & \operatorname{Li}_2(z_g^{\sigma(g)}) - \frac{\pi^2}{6} + \frac{1}{2} \{ \log z_g^{\sigma(g)} + \kappa_g \pi \sqrt{-1} \} \{ \log(1 - z_g^{\sigma(g)}) + \lambda_g \pi \sqrt{-1} \} \\ &= \operatorname{Li}_2(z_g^{\sigma(g)}) - \frac{\pi^2}{6} + \frac{1}{2} u(S_g) \{ v(S_g) + 2 \log(1 - z_g^{\sigma(g)}) \} \end{aligned}$$

modulo π^2 . For simplicity, we put $\hat{v}(S_g) = v(S_g) + 2 \log(1 - z_g^{\sigma(g)})$.

Lemma 3.3. $L(4c+1) + L(4c+2) = 0 = L(4c+3) + L(4c+4)$.

Proof. We prove the first equality. Suppose $\beta(\nu^{-1}(c-b)) = \{l, k\}$ as in Corollary 3.2, and put $\gamma = \xi(A(P_l))$, $\delta = \xi(A(P_k))$. Then, we have $-u(S_{4c+1}) = \log \gamma - \log \delta = u(S_{4c+2})$,

$$\hat{v}(S_{4c+1}) = \log \xi(A(X_{c-b})) - \log \xi(U_+) + \log \gamma - \log \xi(C(X_{c-b})) + 2 \log(1 - \delta/\gamma),$$

$$\hat{v}(S_{4c+2}) = \log \delta - \log \xi(U_+) + \log \xi(A(X_{c-b})) - \log \xi(B(X_{c-b})) + 2 \log(1 - \gamma/\delta),$$

and

$$\begin{aligned} L(4c+1) + L(4c+2) &= \operatorname{Li}_2(\delta/\gamma) + \operatorname{Li}_2(\gamma/\delta) - \frac{\pi^2}{3} + \frac{1}{2} (\log \gamma - \log \delta) \{ \log \delta - \log \gamma \\ &\quad + 2 \log(1 - \gamma/\delta) - 2 \log(1 - \delta/\gamma) \} \end{aligned}$$

by definition. The last term is further equal to

$$\begin{aligned} & \frac{1}{2} \{ \log(-\gamma/\delta) \pm \pi \sqrt{-1} \} (-\log(-\gamma/\delta) \mp \pi \sqrt{-1} + 2 \log(-\gamma/\delta)) \\ &= \frac{1}{2} \{ \log^2(-\gamma/\delta) + \pi^2 \}, \end{aligned}$$

and $L(4c+1) + L(4c+2) = 0$ follows from the dilogarithm identity

$$\operatorname{Li}_2(\delta/\gamma) + \operatorname{Li}_2(\gamma/\delta) = -\frac{1}{2} \log^2(-\gamma/\delta) - \frac{\pi^2}{6}. \quad \square$$

4. Proof of Theorem 2.6

In this section, we prove Theorem 2.6. By (3.1) and Lemma 3.3, we have

$$\text{cv}(M) = \sum_{g \in \Gamma} \sigma(g) L(g) = V(\zeta; D) + \frac{1}{2} \sum_{g \in \Gamma} \sigma(g) u(S_g) \hat{v}(S_g)$$

modulo π^2 . Furthermore, if we put $u(P_l) = \log \xi(A(P_l)) - \log \xi(B(P_l))$ for $l \in \Lambda$,

$$\sum_{g \in \Gamma} \sigma(g) u(S_g) \hat{v}(S_g) = \sum_{g \in \Gamma} \sigma(g) \{u(P_{\beta(g)}) - u(P_{\alpha(g)})\} \hat{v}(S_g) = - \sum_{e=1}^{2(c-a-b)-1} \chi(e),$$

where

$$\chi(e) = \sum_{\epsilon(\alpha(g))=e} \sigma(g) u(P_{\alpha(g)}) \hat{v}(S_g) - \sum_{\epsilon(\beta(g))=e} \sigma(g) u(P_{\beta(g)}) \hat{v}(S_g).$$

Therefore, since $u(P_l) = 0$ if $P_l \in \partial F_0$, we have

$$\text{cv}(M) = V(\zeta; D) - \frac{1}{2} \sum_{e=1}^d \chi(e),$$

and the following completes the proof of Theorem 2.6.

Claim 4.1. Suppose $1 \leq e \leq d$. Then,

$$\frac{1}{2} \chi(e) \equiv \log \zeta(e) \left[z(e) \frac{\partial V(z; D)}{\partial z(e)} \right]_{z=\zeta} \pmod{2\pi^2}. \quad (4.1)$$

Proof. Suppose E_e is surrounded by Q_g, Q_h, Q_i, Q_j as shown in Fig. 14. Note that

$$C(R_{\mu(g)}) = C(R_{\mu(h)}), \quad C(R_{\mu(i)}) = C(R_{\mu(j)}).$$

If $\alpha(g) = \alpha(i)$ and $\alpha(h) = \alpha(j)$, we have $u(P_{\alpha(g)}) = u(P_{\alpha(h)}) = u(P_{\alpha(i)}) = u(P_{\alpha(j)})$ and

$$\chi(e) = u(P_{\alpha(g)}) \{ \hat{v}(S_g) - \hat{v}(S_h) - \hat{v}(S_i) + \hat{v}(S_j) \}.$$

Then, by definition,

$$\begin{aligned} \hat{v}(S_g) - \hat{v}(S_i) &= -\log \xi(C(R_{\mu(g)})) + \log \xi(C(R_{\mu(i)})) \\ &\quad + 2 \log(1 - z_g) - 2 \log(1 - z_i^{-1}), \\ \hat{v}(S_h) - \hat{v}(S_j) &= -\log \xi(C(R_{\mu(h)})) + \log \xi(C(R_{\mu(j)})) \\ &\quad + 2 \log(1 - z_h^{-1}) - 2 \log(1 - z_j), \end{aligned}$$

and (4.1) follows from Lemma 3.1. The case $\beta(g) = \beta(i)$, $\beta(h) = \beta(j)$ is quite similar. If $\alpha(g) = \alpha(i)$ and $\beta(h) = \beta(j)$, we have $u(P_{\alpha(g)}) = u(P_{\beta(h)}) = u(P_{\alpha(i)}) =$

$u(P_{\beta(j)})$ and

$$\chi(e) = u(P_{\alpha(g)})\{\hat{v}(S_g) - \hat{v}(S_h) - \hat{v}(S_i) + \hat{v}(S_j)\}.$$

Then, by definition,

$$\begin{aligned}\hat{v}(S_g) - \hat{v}(S_i) &= -\log \xi(C(R_{\mu(g)})) + \log \xi(C(R_{\mu(i)})) \\ &\quad + 2\log(1 - z_g) - 2\log(1 - z_i^{-1}), \\ \hat{v}(S_h) - \hat{v}(S_j) &= -\log \xi(C(R_{\mu(j)})) + \log \xi(C(R_{\mu(h)})) \\ &\quad + 2\log(1 - z_h) - 2\log(1 - z_j^{-1}),\end{aligned}$$

and (4.1) follows from Lemma 3.1. This completes the proof. $\square$

5. Example

Suppose D with P_0 removed looks like Fig. 21.

The subgraph G of D and the edges $E_1, E_2, \dots, E_9$ of G are given in Fig. 22.

For simplicity, we put $w_e = z(e)$ for $1 \leq e \leq 5$. Then, $V(z; D)$ is given by

$$\begin{aligned}&\text{Li}_2(w_1/w_4) - \text{Li}_2(w_1/w_3) + \text{Li}_2(w_1) - \text{Li}_2(1/w_4) + \text{Li}_2(w_2/w_4) - \text{Li}_2(w_2) \\ &\quad - \text{Li}_2(1/w_2) + \text{Li}_2(w_5/w_2) - \text{Li}_2(w_5) - \text{Li}_2(1/w_5) \\ &\quad + \text{Li}_2(w_3/w_5) - \text{Li}_2(w_3) + \pi^2/3\end{aligned}$$

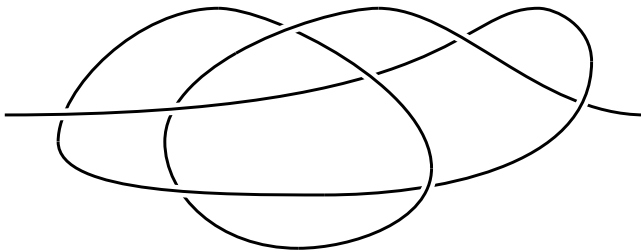


Fig. 21.

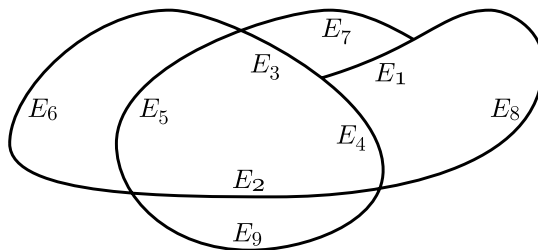


Fig. 22.

and the hyperbolicity equations for M are given by

$$\begin{aligned}\exp\left\{w_1\frac{\partial V}{\partial w_1}\right\} &= \frac{1-w_1/w_3}{(1-w_1/w_4)(1-w_1)} = 1, \\ \exp\left\{w_2\frac{\partial V}{\partial w_2}\right\} &= \frac{(1-w_2)(1-w_5/w_2)}{(1-w_2/w_4)(1-1/w_2)} = 1, \\ \exp\left\{w_3\frac{\partial V}{\partial w_3}\right\} &= \frac{1-w_3}{(1-w_1/w_3)(1-w_3/w_5)} = 1, \\ \exp\left\{w_4\frac{\partial V}{\partial w_4}\right\} &= \frac{(1-w_1/w_4)(1-w_2/w_4)}{1-1/w_4} = 1, \\ \exp\left\{w_5\frac{\partial V}{\partial w_5}\right\} &= \frac{(1-w_5)(1-w_3/w_5)}{(1-w_5/w_2)(1-1/w_5)} = 1.\end{aligned}$$

There are five essential solutions $z = \zeta_1, \zeta_2, \zeta_3, \zeta_4, \zeta_5$ to these equations characterized by

$$\begin{aligned}\widehat{V}(\zeta_1; D) &= (11.9099\dots) + (4.1249\dots)\sqrt{-1}, \\ \widehat{V}(\zeta_2; D) &= (11.9099\dots) - (4.1249\dots)\sqrt{-1}, \\ \widehat{V}(\zeta_3; D) &= (1.8513\dots) + (1.1089\dots)\sqrt{-1}, \\ \widehat{V}(\zeta_4; D) &= (1.8513\dots) - (1.1089\dots)\sqrt{-1}, \\ \widehat{V}(\zeta_5; D) &= -1.2036\dots \in \mathbb{R},\end{aligned}$$

whose imaginary parts represent the volumes of the corresponding parabolic representations of $\pi_1(M)$, and we have $\text{cv}(M) = \widehat{V}(\zeta_1; D)$.

Acknowledgments

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